

- 3 In 2014, scientists published a phylogenetic (evolutionary) tree diagram showing how and when different groups of insects developed from ancestral groups.

To construct the diagram, the scientists compared nucleotide **and** amino acid sequence data for 1478 genes from each of 103 different insect species. The 103 species represented all 32 orders of the class Insecta.

The data were used to arrange the species in related groups according to their degree of molecular similarity and to track how sequences changed over time.

- (a) (i) Outline **three** ways in which the members of a species are similar.

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..... [3]

- (ii) Organisms are classified in a hierarchical system of taxonomic units.

Name the domain and kingdom to which insects belong.

domain

kingdom

[2]

- (iii) Explain why a phylogenetic tree diagram is useful for classifying insect species.

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..... [2]

(b) For this study, new software for processing large quantities of data was developed.

- (i)** Calculate the total number of sequences that were compared, considering both the nucleotide and amino acid sequences for each gene.

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- (ii)** Explain why this study needs to make use of bioinformatic techniques.

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..... [3]

[Total: 11]